

Glycolysis

Jason Ryan, MD, MPH

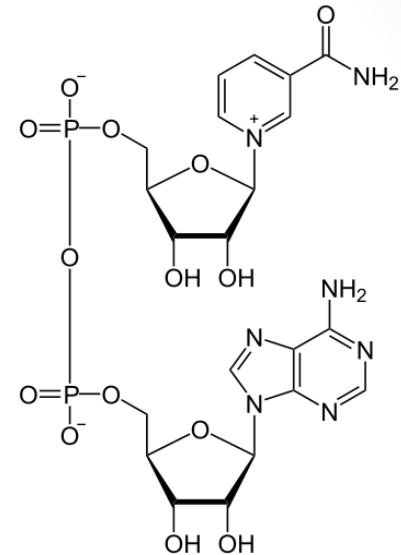
Glycolysis

- Used by all cells of the body
- Sequence of reactions that occurs in **cytoplasm**
- Converts **glucose** (6 carbons) to **pyruvate** (3 carbons)
- Generates ATP and NADH

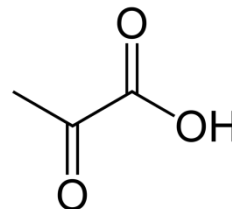
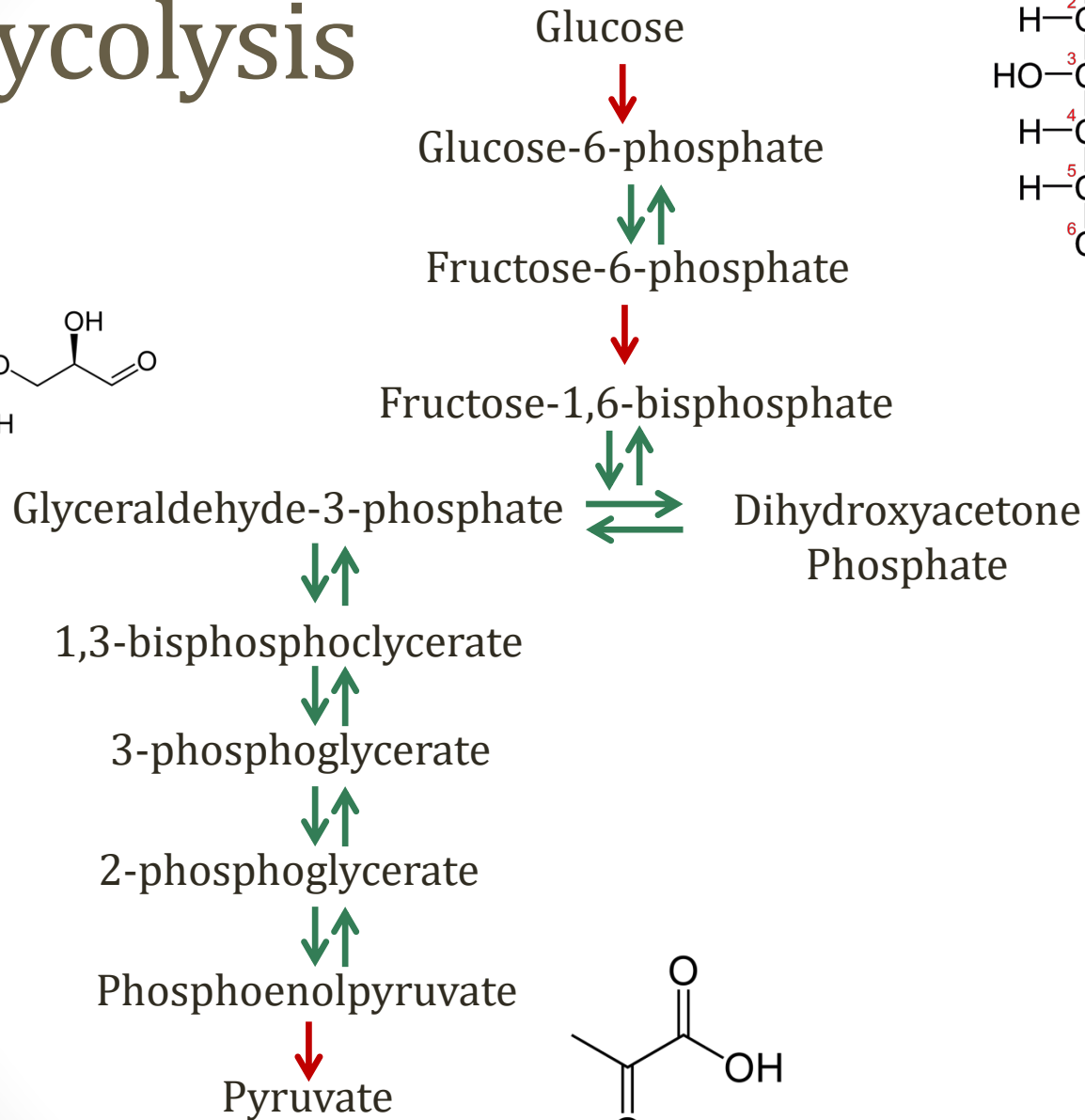
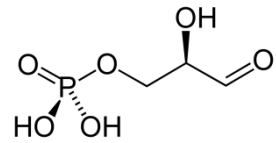
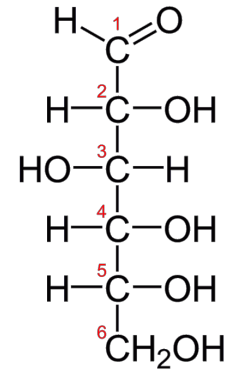
NADH

Nicotinamide adenine dinucleotide

- Two nucleotides
- Carries electrons
- **NAD⁺**
 - Accepts electrons
- **NADH**
 - Donates electrons
 - Can donate to electron transport chain → ATP



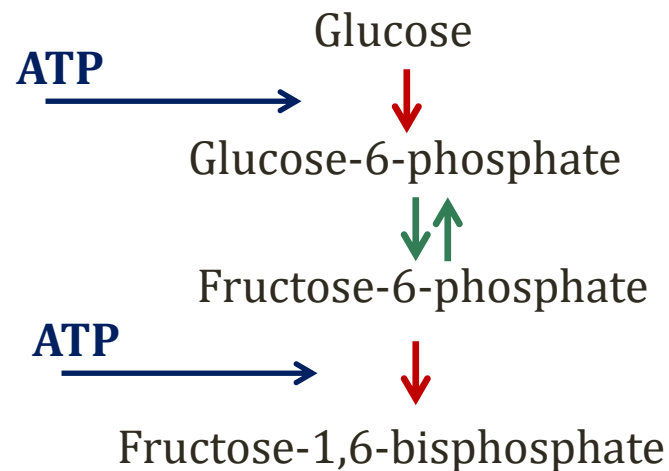
Glycolysis



Glycolysis

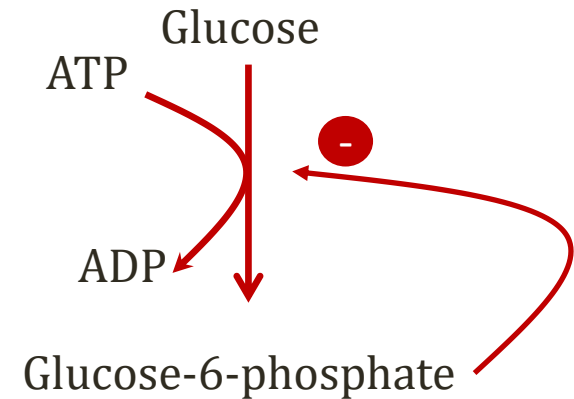
Priming Stage

- Uses energy (consumes 2 ATP)
- First and last reactions most critical



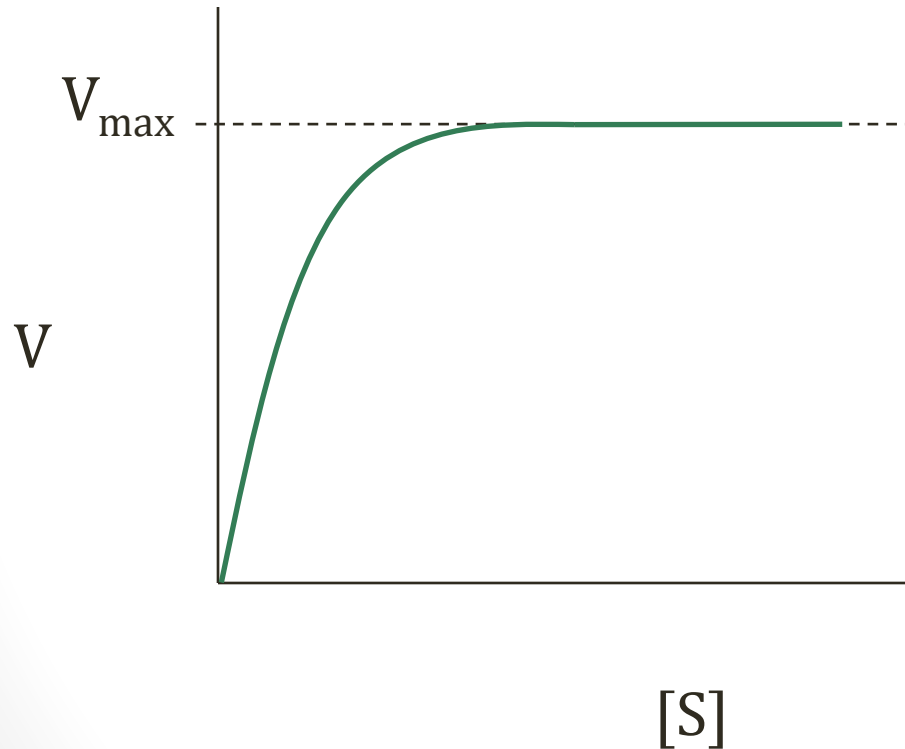
Hexokinase vs. Glucokinase

- Hexokinase
 - Found in most tissues
 - Strongly inhibited by G6P
 - Blocks cells from hording glucose
 - Insulin = no effect
 - Low K_m (usually operates max)
 - Low V_m (max is not that high)



Hexokinase

$$V = \frac{V_m * [S]}{K_m + [S]}$$



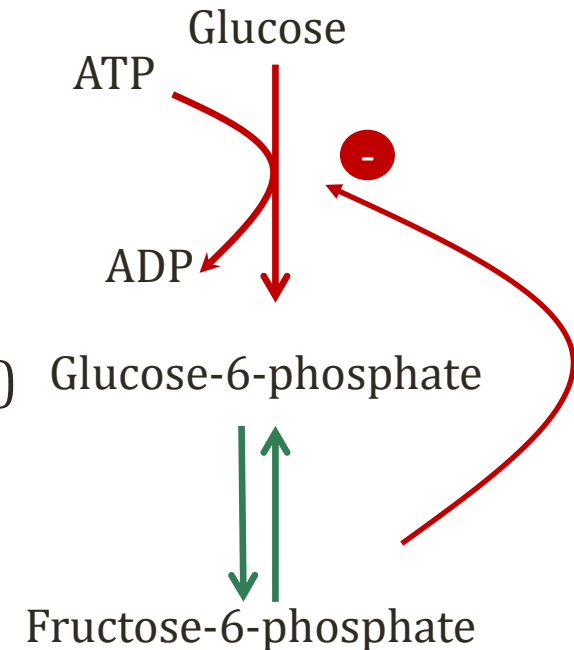
Hexokinase

Low K_m
Quickly Reach V_m
V_m low

Hexokinase vs. Glucokinase

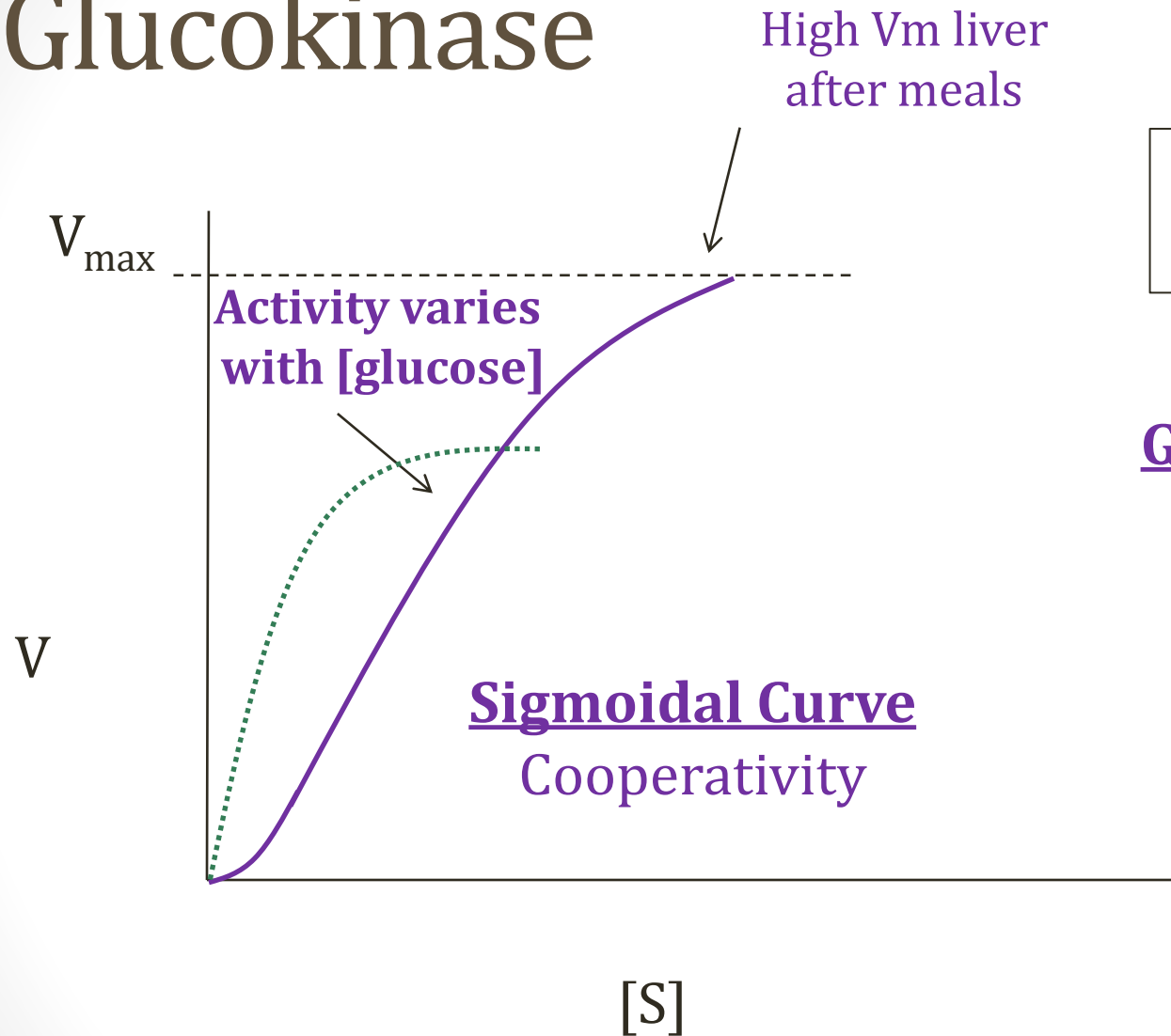
- Glucokinase

- Found in liver and pancreas
- NOT inhibited by G6P
- Induced by insulin
- Insulin promotes transcription
- Inhibited by F6P (overcome by ↑glucose)
- High K_m (rate varies with glucose)



*Enzyme inactive when (1) low glucose and (2) high F6P

Glucokinase



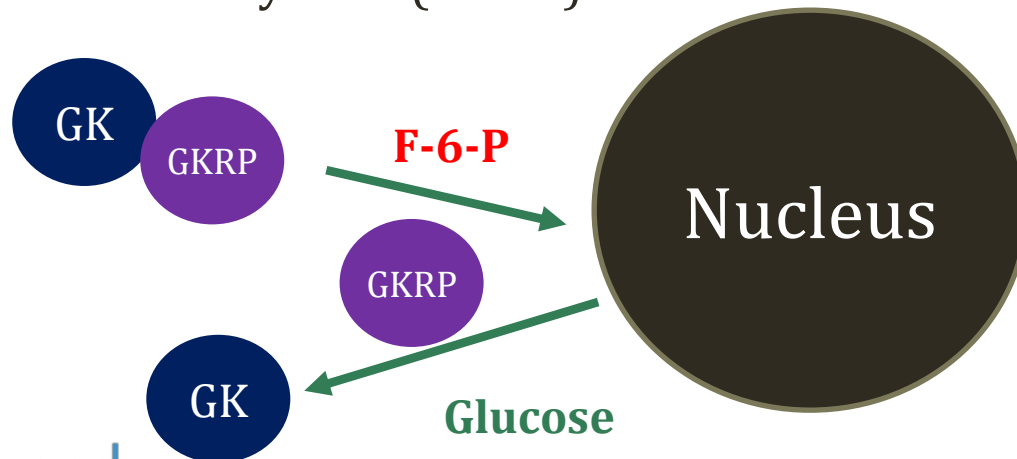
$$V = \frac{V_m * [S]}{K_m + [S]}$$

Glucokinase

High K_m
High V_m

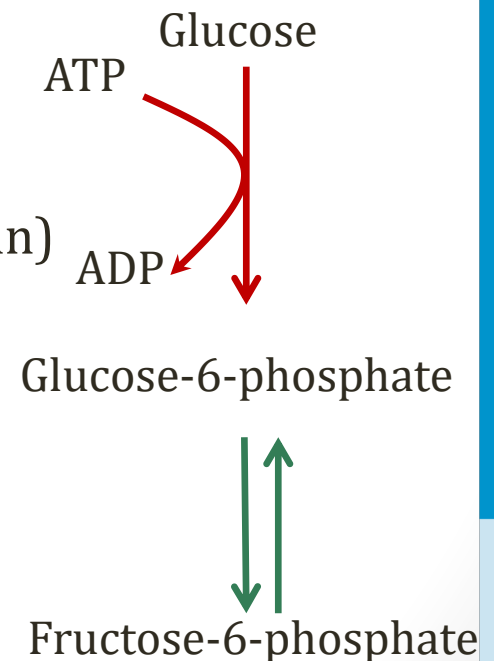
Glucokinase regulatory protein (GKRP)

- Translocates glucokinase to nucleus
- Result: inactivation of enzyme
- Fructose 6 phosphate:
 - GKRP binds glucokinase → nucleus (inactive)
- Glucose:
 - Competes with GKRP for GK binding
 - Glucokinase → cytosol (active)



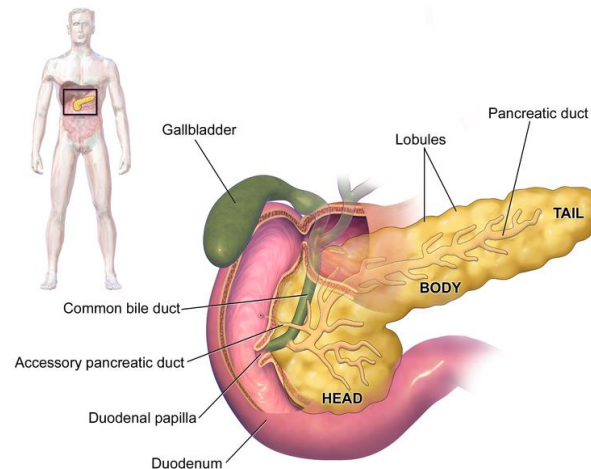
Hexokinase vs. Glucokinase

- **Low blood sugar**
 - Hexokinase working (no inhibition G6P)
 - Glucokinase inactive (rate $\propto$ glucose; low insulin)
 - Glucose to tissues, not liver
- **High blood sugar**
 - Hexokinase inactive (inhibited by G6P)
 - Glucokinase working (high glucose, high insulin)
 - Liver will store glucose as glycogen



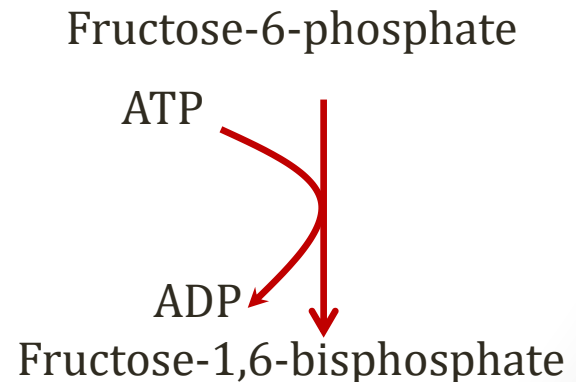
Glucokinase Deficiency

- Results in **hyperglycemia**
- **Pancreas** less sensitive to glucose
- Mild hyperglycemia
- Often **exacerbated by pregnancy**



Phosphofructokinase-1

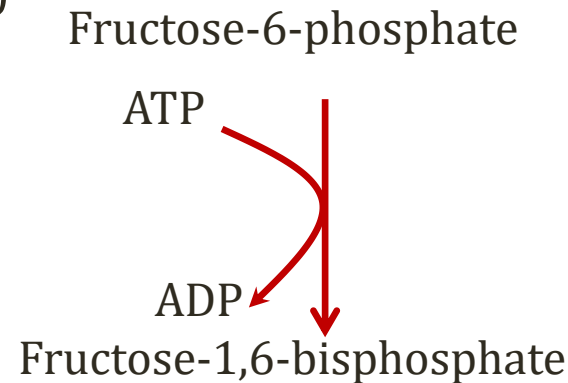
- **Rate limiting step** for glycolysis
- Consumes 2nd ATP in priming stage
- Irreversible
- Commits glucose to glycolysis
 - HMP shunt, glycogen synthesis no long possible



Regulation of Glycolysis

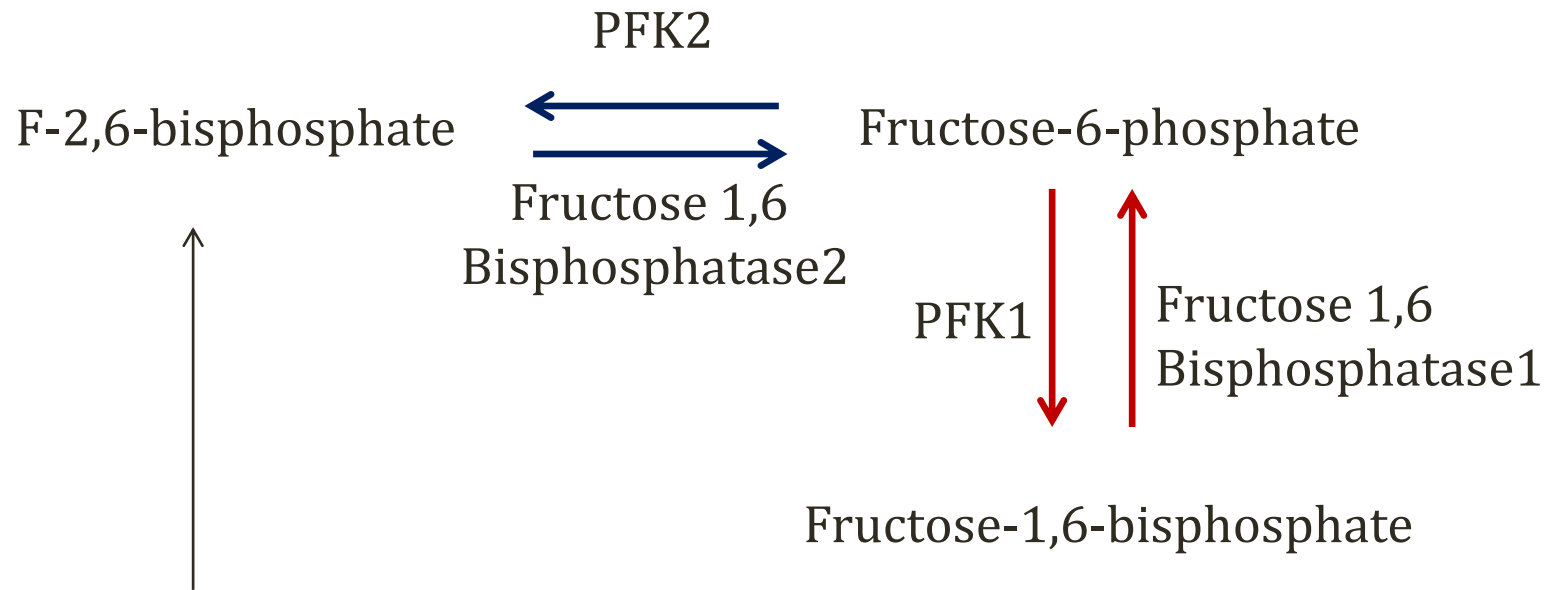
Phosphofructokinase-1

- Key **inhibitors** (less glycolysis)
 - Citrate (TCA cycle)
 - ATP
- Key **inducers** (more glycolysis)
 - AMP
 - Fructose 2,6 bisphosphate (insulin)



Fructose 2,6 Bisphosphate

Regulation of Glycolysis

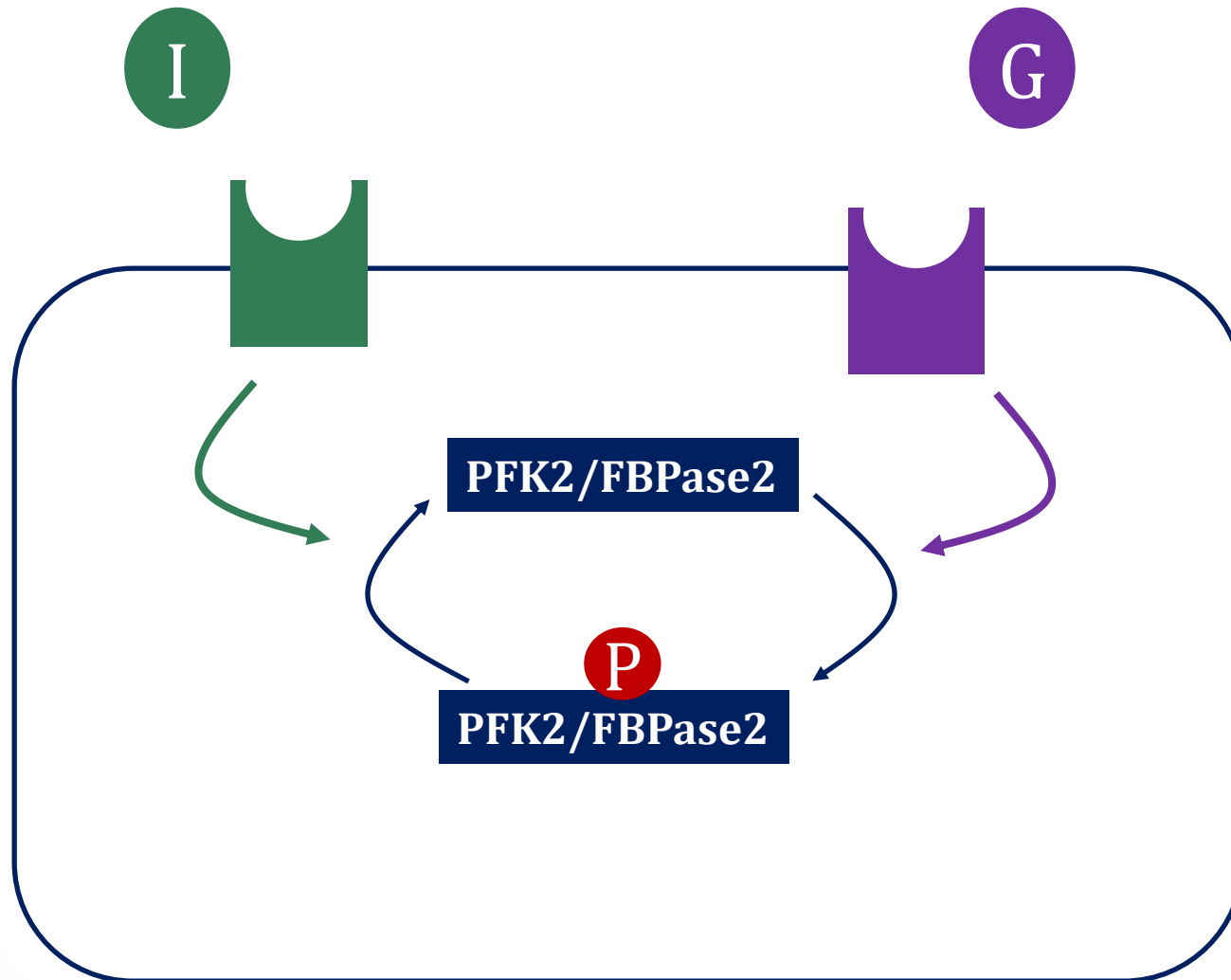


On/off switch glycolysis

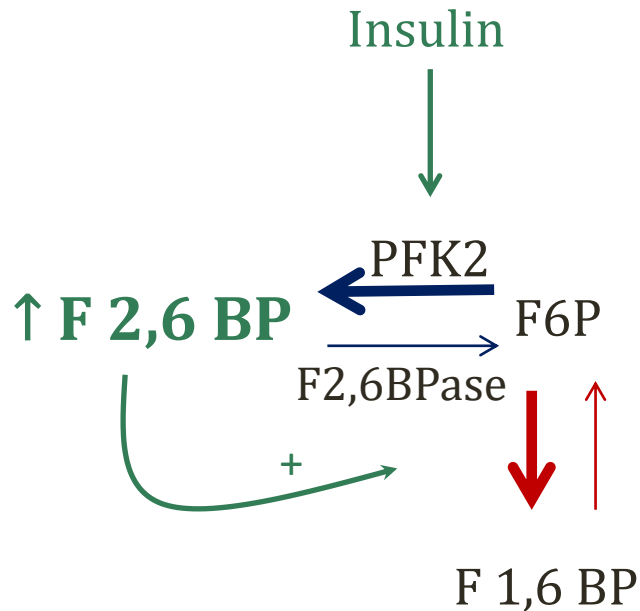
↑ = glycolysis (on)

↓ = no glycolysis (gluconeogenesis)

Insulin and Glucagon



Regulation of Glycolysis



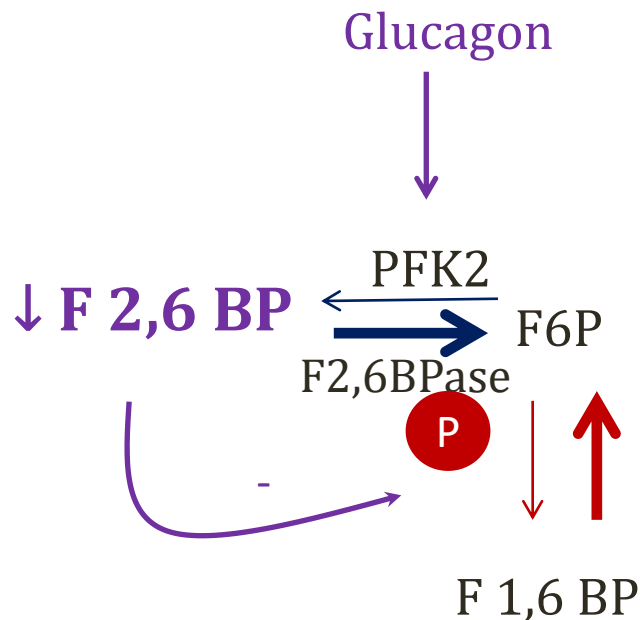
Fed State

↑Insulin
↑F 2,6 BP



Fructose 2,6 Bisphosphate

Regulation of Glycolysis



Fasting State

↓ Insulin
↓ F 2,6 BP

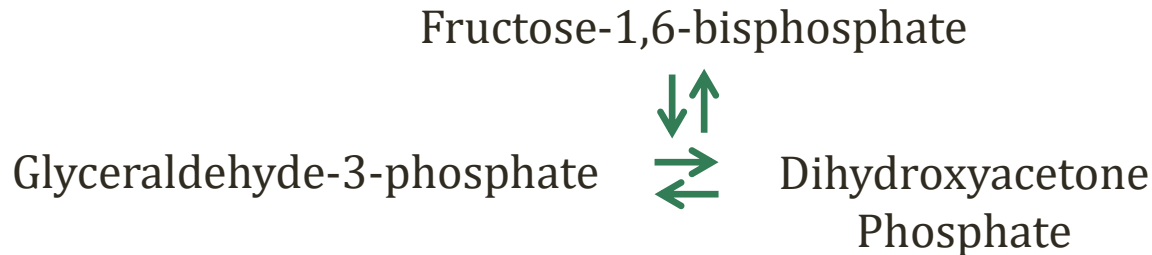


Aude/Wikipedia

Glycolysis

Splitting Stage

- Fructose 1,6-phosphate to two molecules GAP
- Reversible for gluconeogenesis

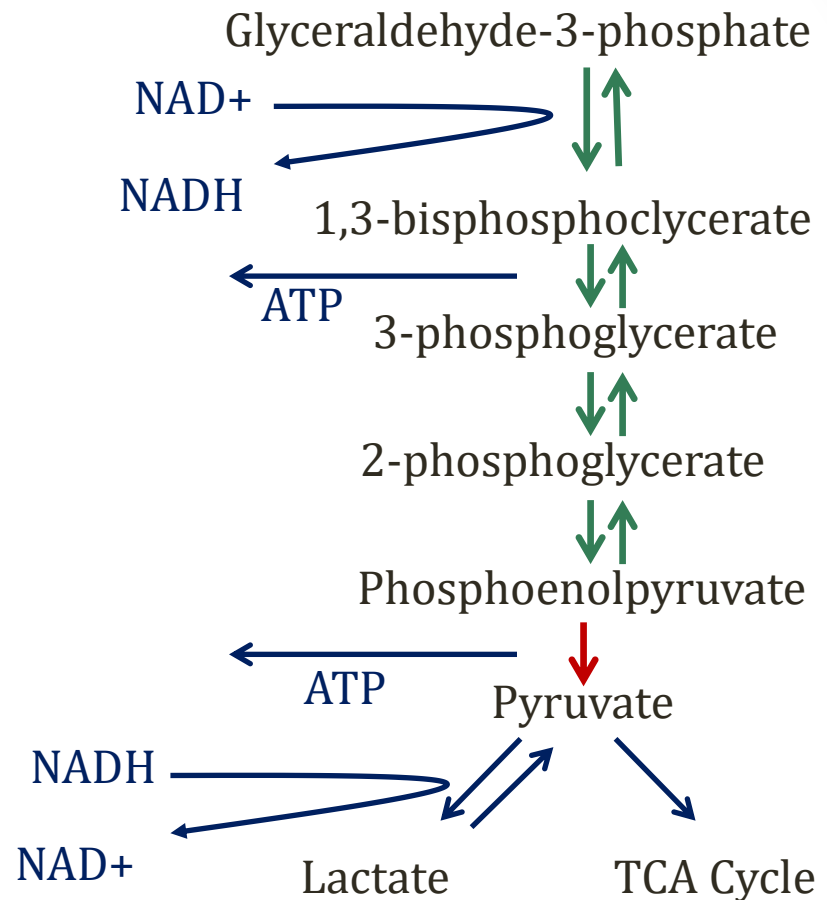


Glycolysis

Energy Stage

- Starts with GAP
- Two ATP per GAP
- Total per glucose = 4

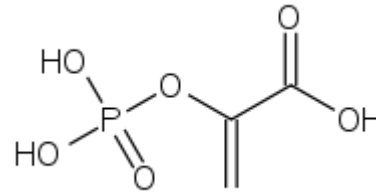
Anaerobic Metabolism (no O₂)
2 ATP (net)



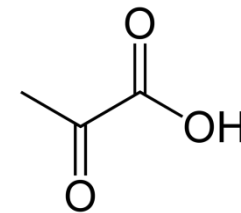
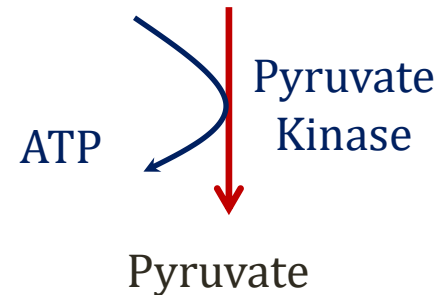
Glycolysis

Energy Stage

- Pyruvate kinase
 - Not reversible
- Inhibited by **ATP, alanine**
- Activated by fructose 1,6 BP
 - “Feed forward” activation
- Glucagon/epinephrine
 - Phosphorylation
 - Inactivation of pyruvate kinase
 - Slows glycolysis/favors gluconeogenesis

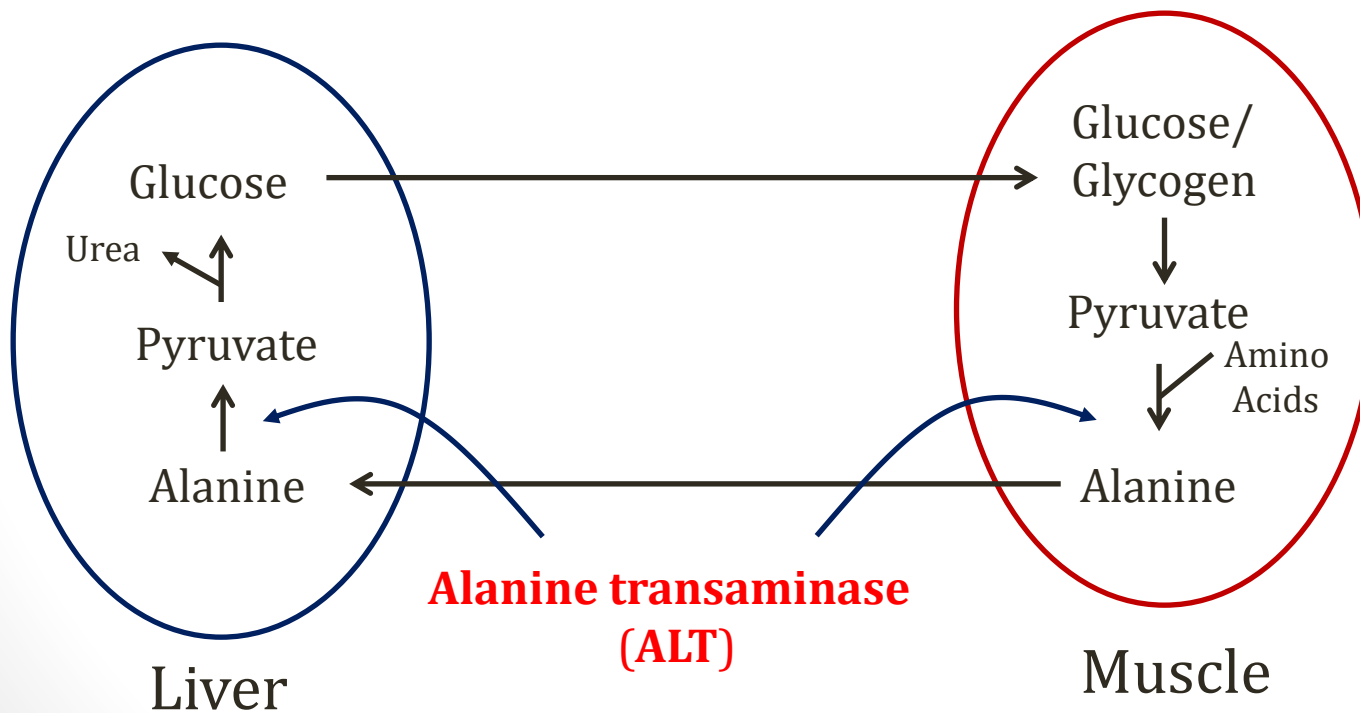


Phosphoenolpyruvate



Alanine Cycle

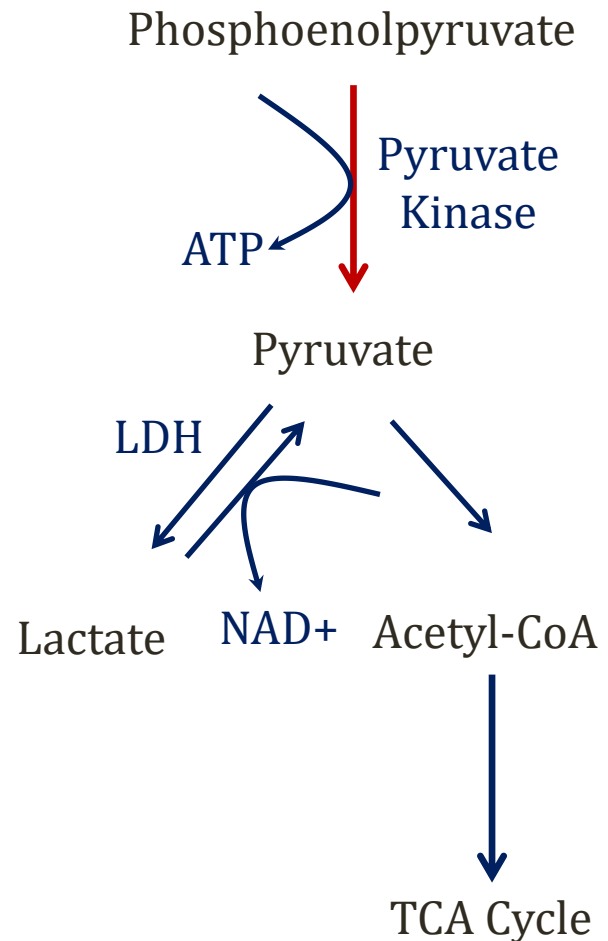
- Skeletal muscles can degrade protein for energy
- Produce **alanine** → blood → liver
- Liver converts alanine to glucose



Glycolysis

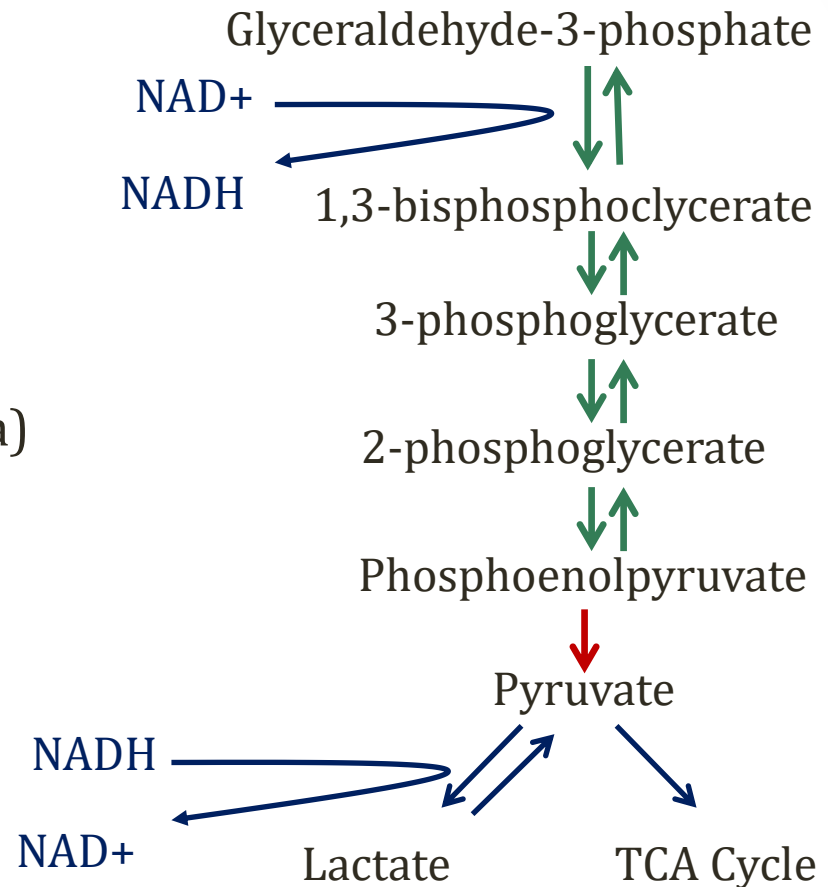
Energy Stage

- Lactate dehydrogenase (LDH)
 - Pyruvate $\leftrightarrow$ Lactate
- Plasma elevations common
 - Hemolysis
 - Myocardial infarction
 - Some tumors
- Pleural effusions
 - Transudate vs. exudate



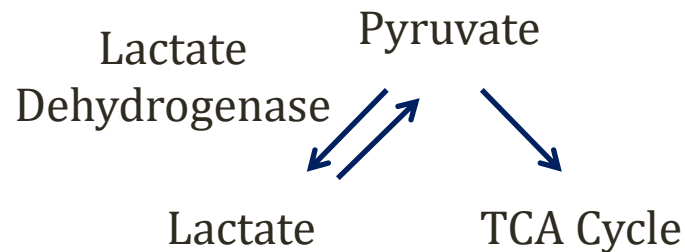
NADH

- Limited supply NAD^+
- Must regenerate
- O_2 present
 - $\text{NADH} \rightarrow \text{NAD}$ (mitochondria)
- O_2 absent
 - $\text{NADH} \rightarrow \text{NAD}^+$ via LDH



Lactic Acidosis

- $\downarrow O_2 \rightarrow \downarrow$ pyruvate entry into TCA cycle
- $\uparrow$ lactic acid production
- $\downarrow pH, \downarrow HCO_3^-$
- Elevated anion gap acidosis
- Sepsis, bowel ischemia, seizures



Muscle Cramps

- Too much exercise → too much NAD consumption
 - Exceed capacity of TCA cycle/electron transport
 - Elevated NADH/NAD ratio
- Favors pyruvate → **lactate**
- pH falls in muscles → cramps
- Distance runners: lots of mitochondria (bigger, too)

Pyruvate Kinase Deficiency

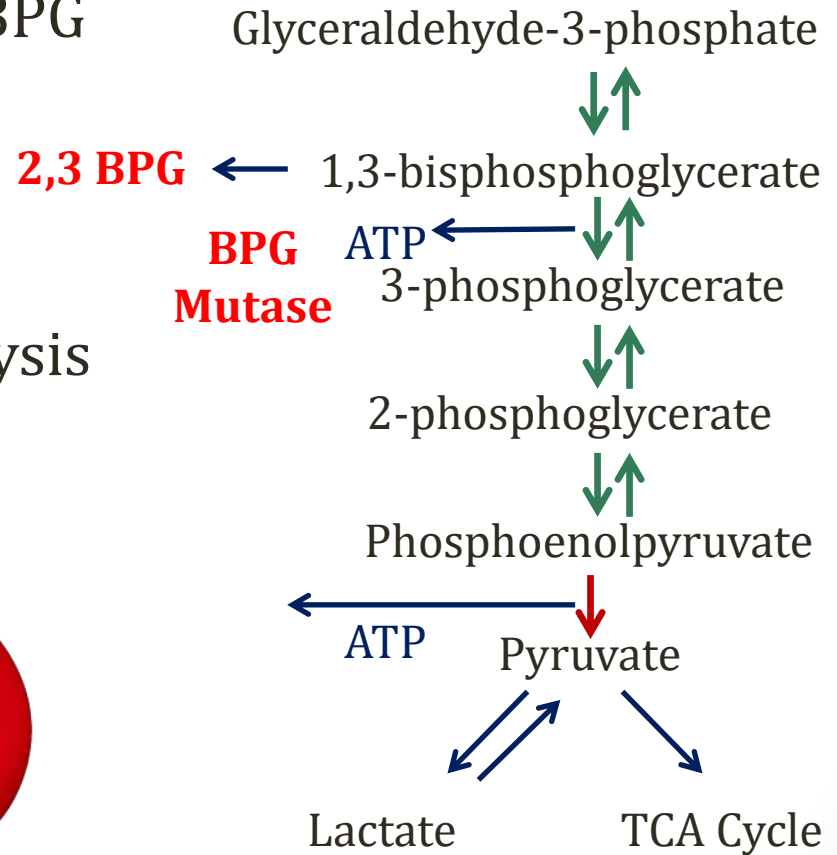
- Autosomal recessive disorder
- **RBCs** most effected
 - No mitochondria
 - Require PK for anaerobic metabolism
 - Loss of ATP
 - Membrane failure → phagocytosis in spleen
- Usually presents as **newborn**
- **Extravascular hemolysis**
- Splenomegaly
- Disease severity ranges based on enzyme activity



Database Center for Life Science (DBCLS)

2,3 Bisphosphoglycerate

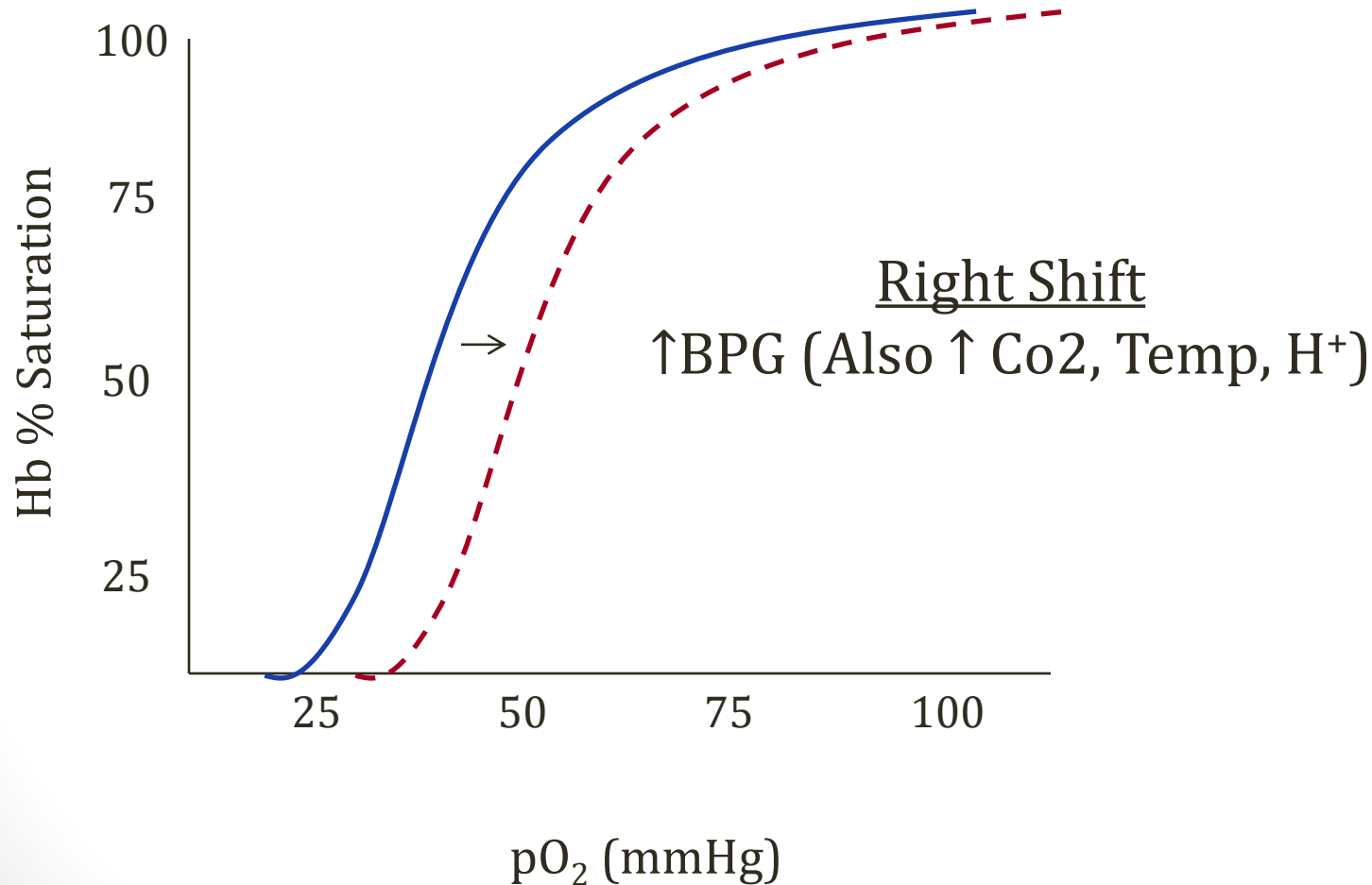
- Created from diverted 1,3 BPG
- Used by **RBCs**
 - No mitochondria
 - No TCA cycle
- **Sacrifices ATP** from glycolysis
- 2,3 BPG alters Hgb binding



Database Center for Life Science (DBCLS)

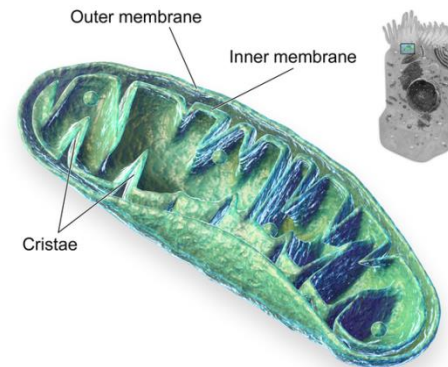
Right Curve Shifts

Easier to release O_2



Energy Yield from Glucose

- ATP generated depends on cells/oxygen
- Highest yield with **O₂ and mitochondria**
 - Allows pyruvate to enter TCA cycle
 - Converts pyruvate/NADH → ATP



Mitochondria

Energy from Glucose

Oxygen and Mitochondria



32 ATP = malate-aspartate shuttle (liver, heart)

30 ATP = glycerol-3-phosphate shuttle (muscle)

No Oxygen or No Mitochondria

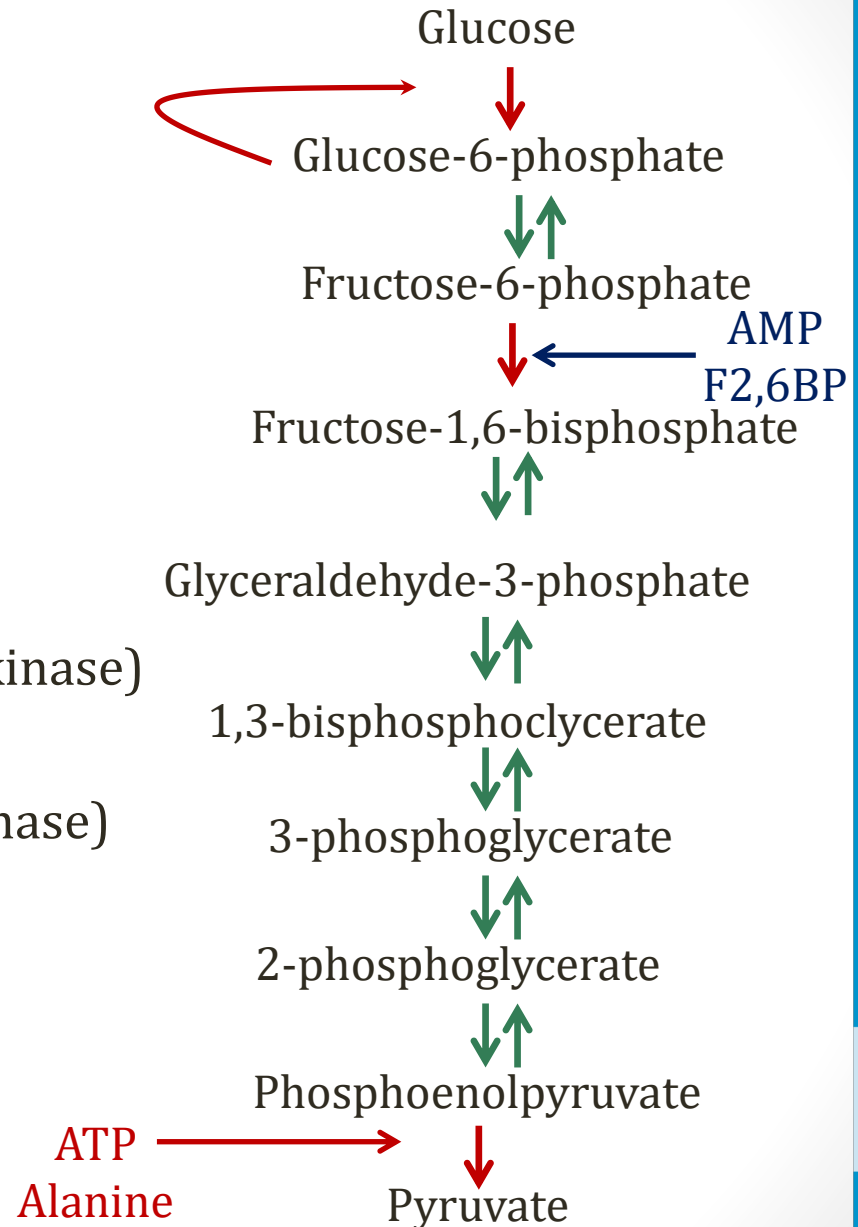


*RBCs = no mitochondria

Summary

Key Steps

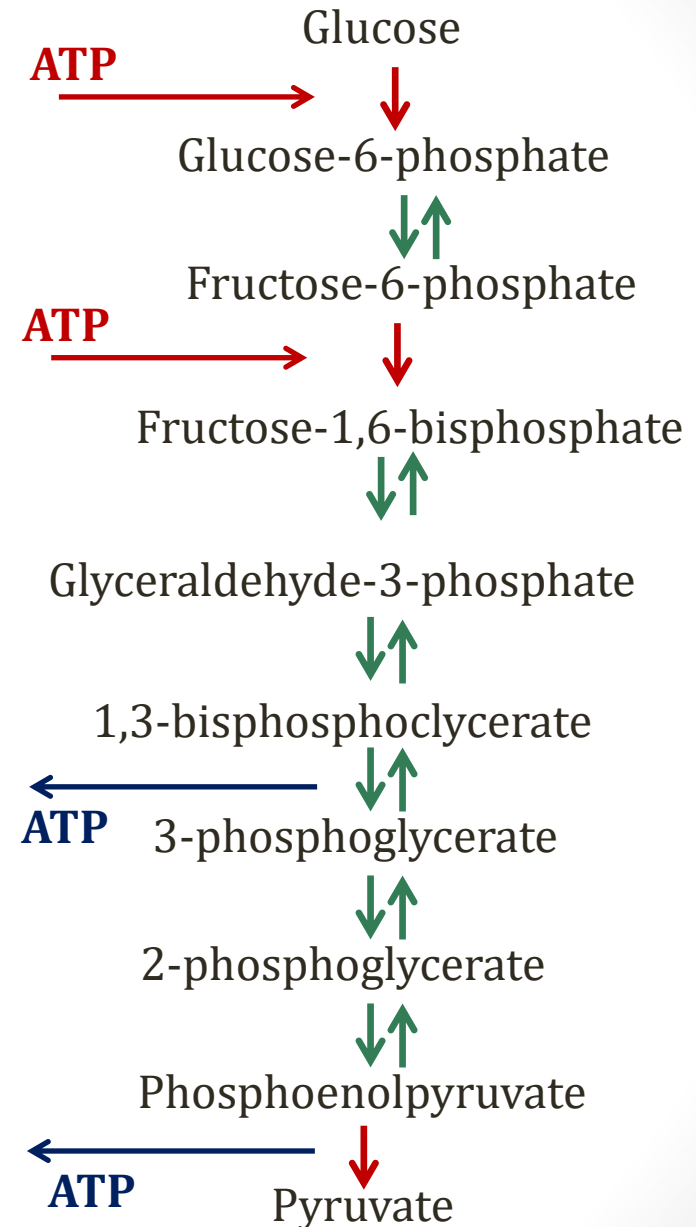
- Regulation
 - #1: Hexokinase/Glucokinase
 - #2: PFK1
 - #3: Pyruvate Kinase
- Irreversible
 - Glucose $\rightarrow$ G6P (Hexo/Glucokinase)
 - F6P $\rightarrow$ F 1,6 BP (PFK1)
 - PEP $\rightarrow$ pyruvate (pyruvate kinase)



Summary

Key Steps

- **ATP expended**
 - Glucose $\rightarrow$ G6P
 - F6P $\rightarrow$ F1,6BP
- **ATP generated**
 - 1,3BPG $\rightarrow$ 3PG
 - PEP $\rightarrow$ pyruvate

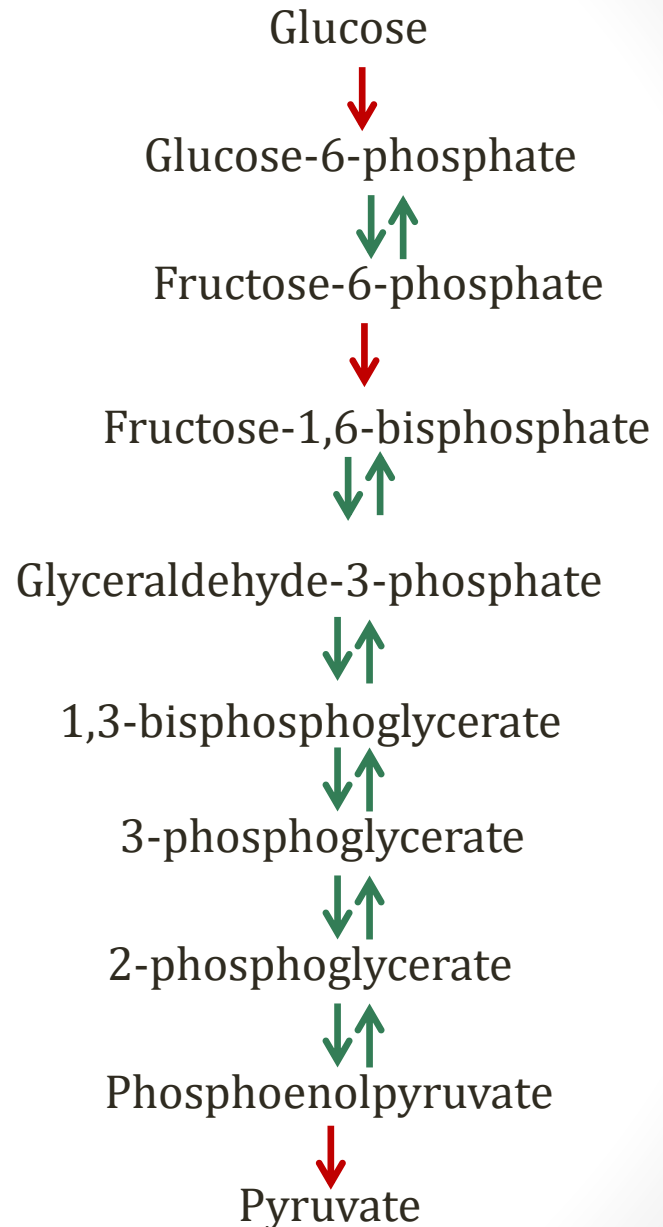


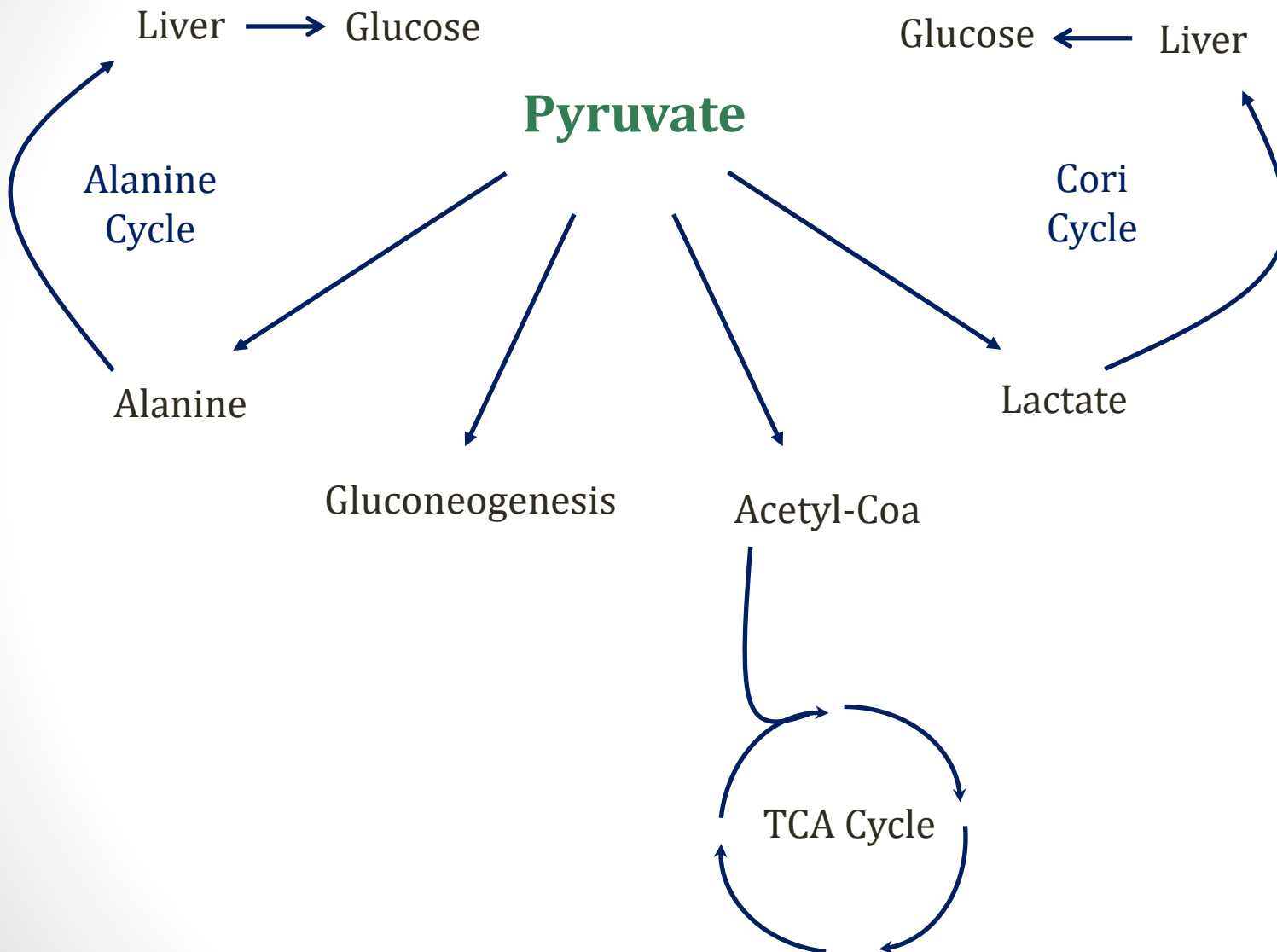
Gluconeogenesis

Jason Ryan, MD, MPH

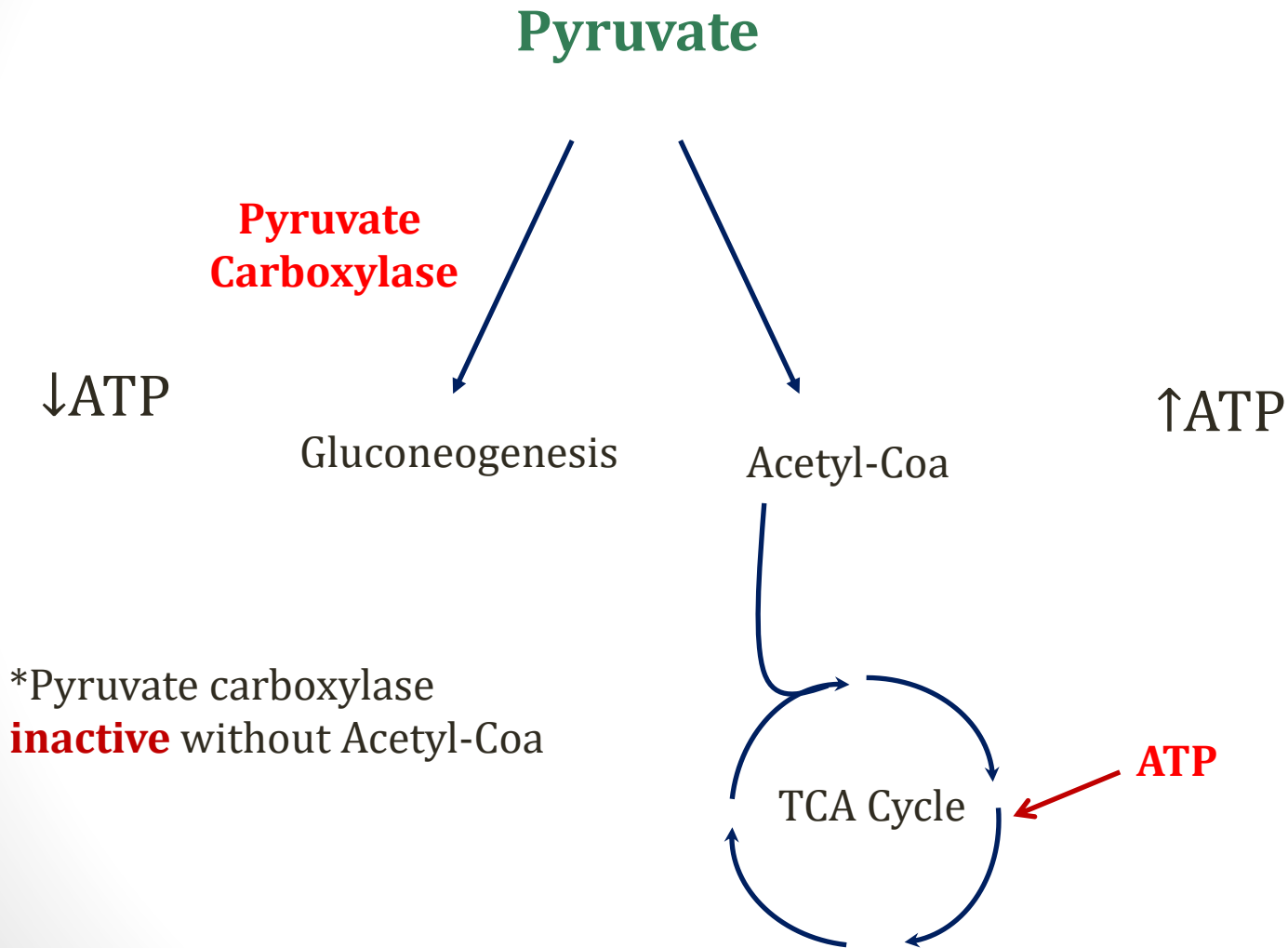
Gluconeogenesis

- Glucose from other carbons
- Sources of glucose
 - Pyruvate
 - Lactate
 - Amino acids
 - Propionate (odd chain fats)
 - Glycerol (fats)



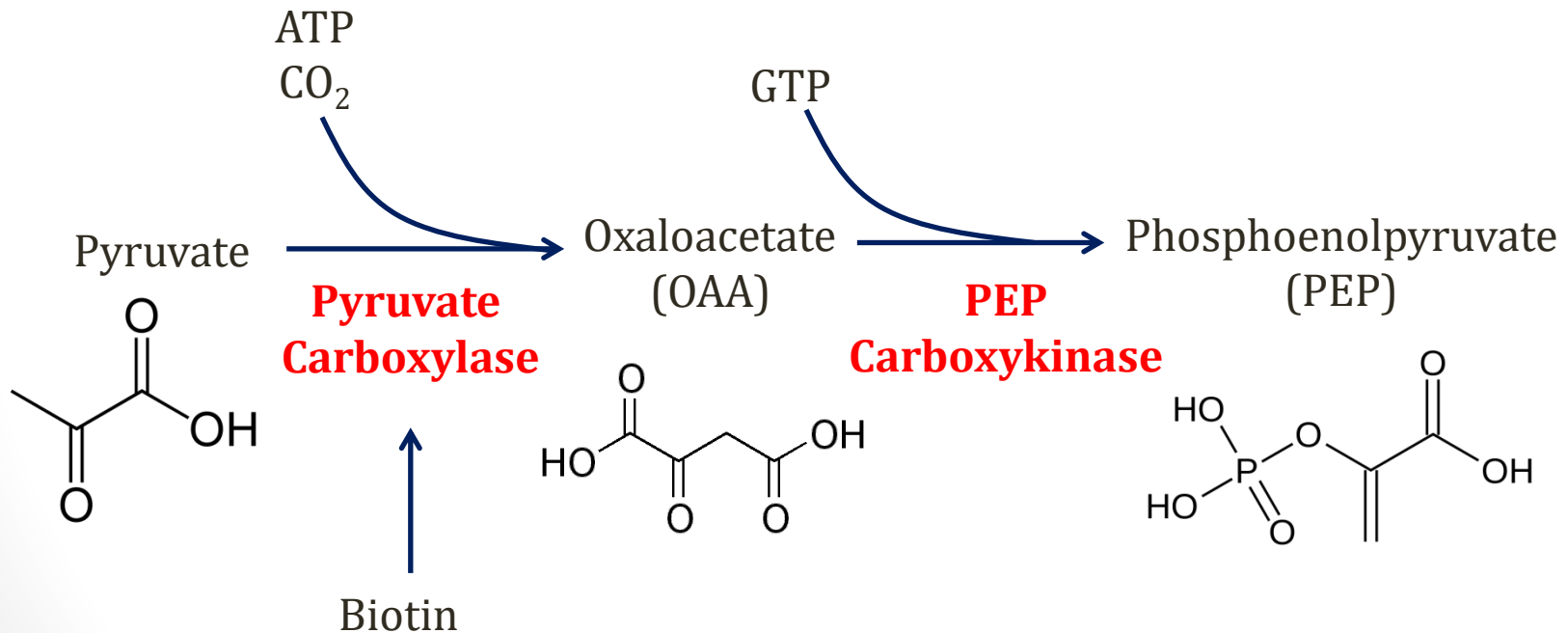


Pyruvate



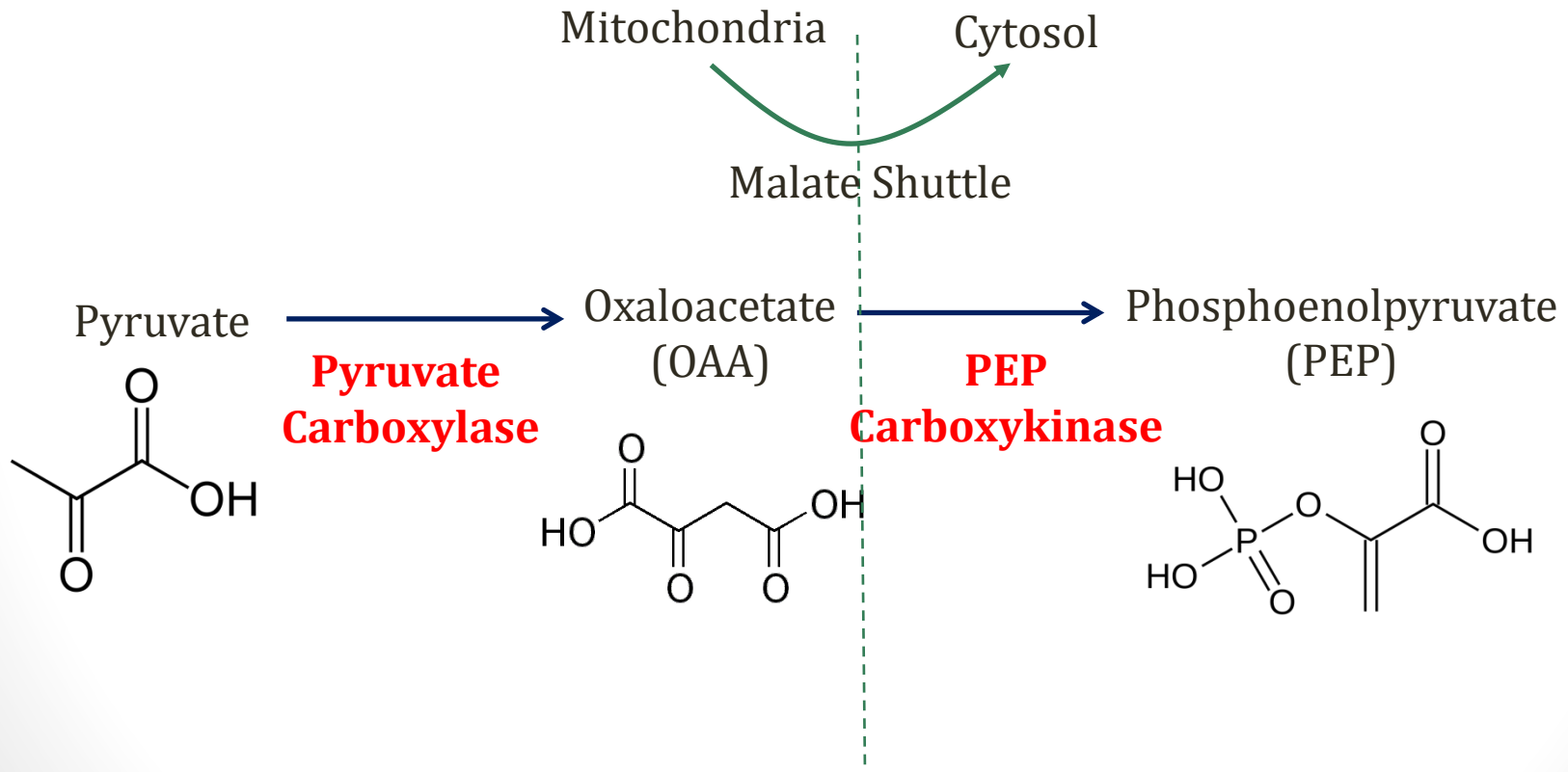
Gluconeogenesis

- Step #1: Pyruvate → Phosphoenolpyruvate

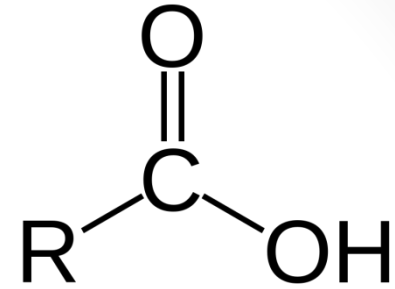


Gluconeogenesis

- Step #1: Pyruvate → Phosphoenolpyruvate



Biotin



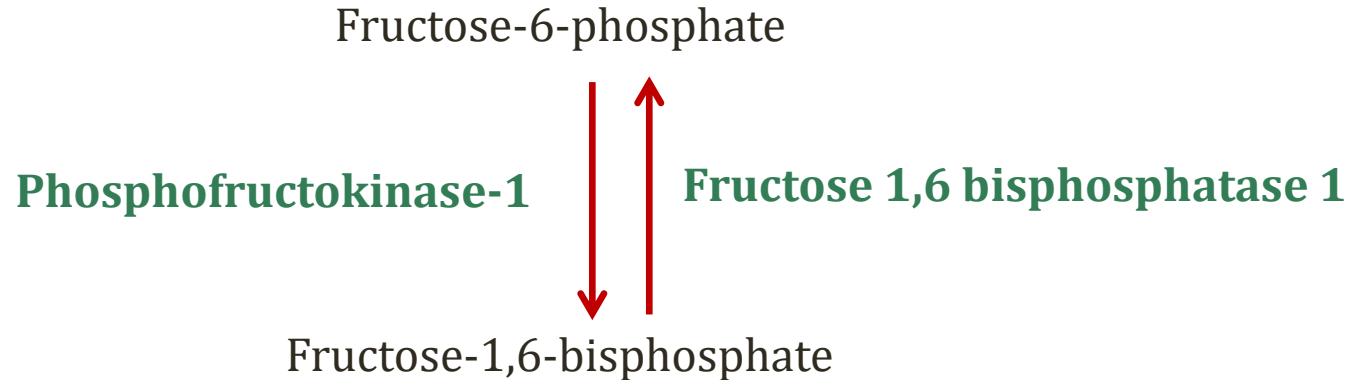
- Cofactor for carboxylation enzymes
 - All add 1-carbon group via CO₂
 - **Pyruvate carboxylase**
 - Acetyl-CoA carboxylase
 - Propionyl-CoA carboxylase
- Deficiency
 - Very rare (vitamin widely distributed)
 - Massive consumption raw egg whites (avidin)
 - Dermatitis, glossitis, loss of appetite, nausea

Pyruvate Carboxylase Deficiency

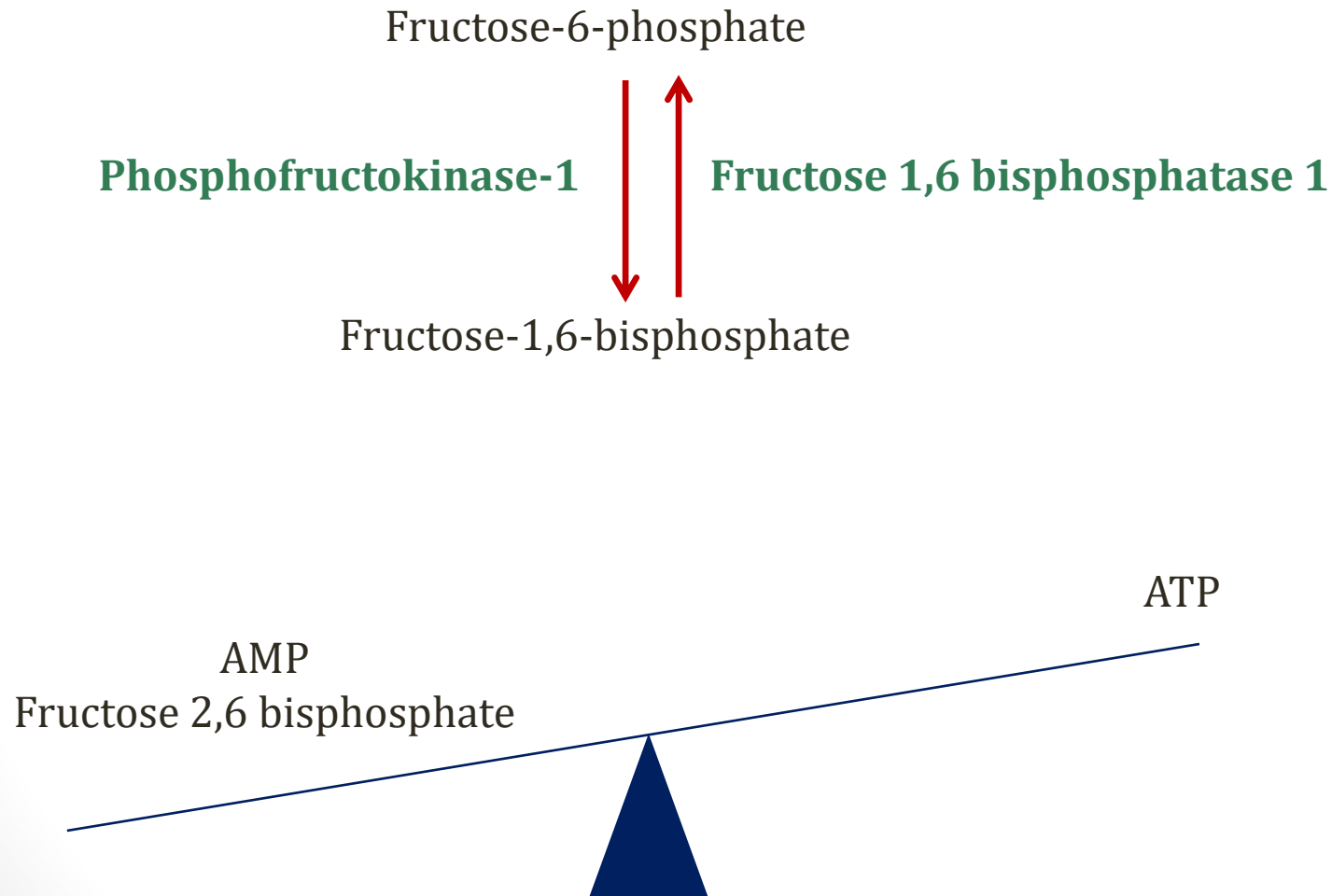
- Very rare
- Presents in infancy with failure to thrive
- Elevated pyruvate → lactate
- Lactic acidosis

Gluconeogenesis

- Step #2:
 - Fructose 1,6 bisphosphate → Fructose 6 phosphate
 - Rate limiting step

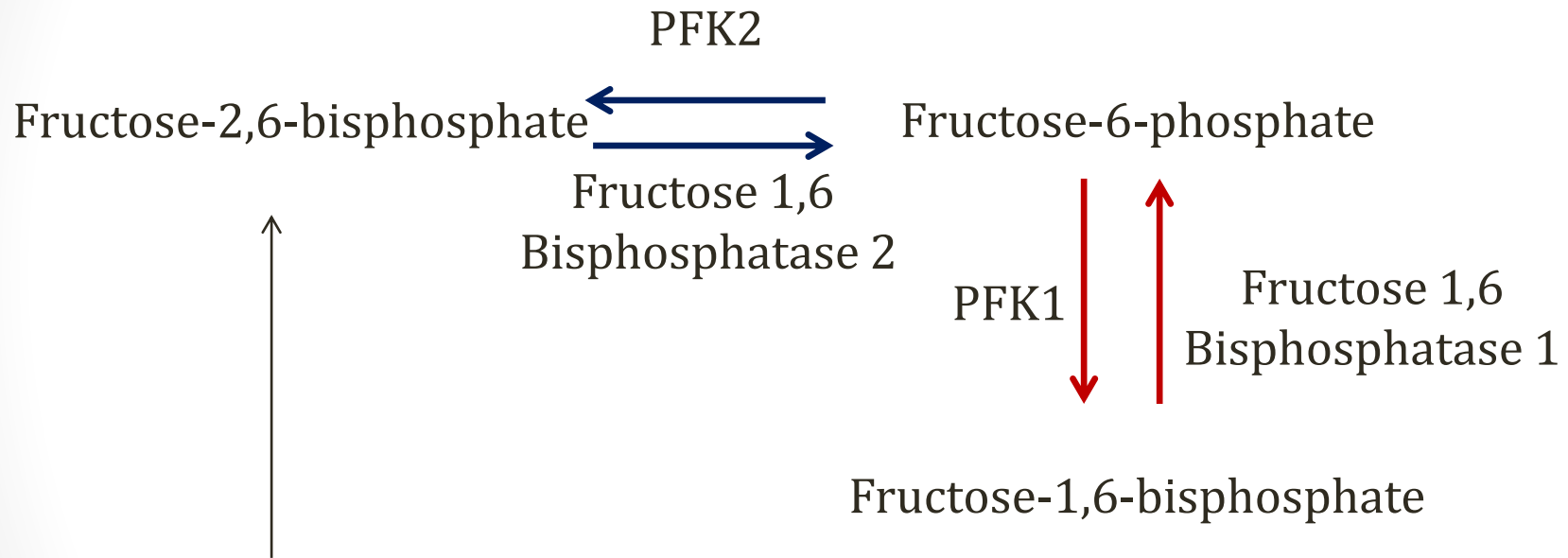


Gluconeogenesis



Fructose 2,6 Bisphosphate

Regulation of Glycolysis/Gluconeogenesis



On/off switch glycolysis

↑ = glycolysis (on)

↓ = no glycolysis (gluconeogenesis)

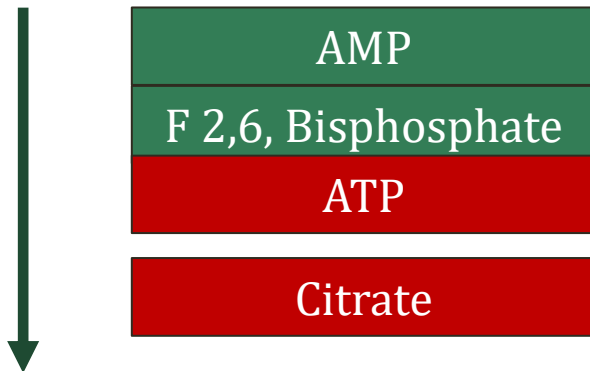
Fructose 2,6 Bisphosphate

Regulation of Gluconeogenesis

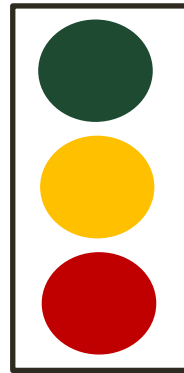
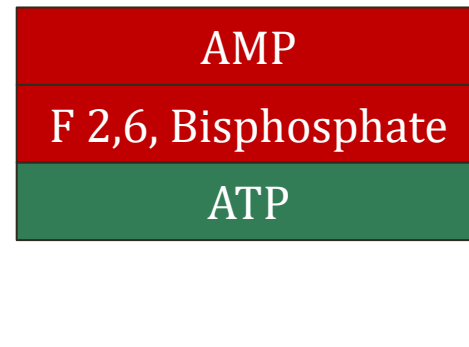
- Levels rise with high insulin (fed state)
- Levels fall with high glucagon (fasting state)
- Drives glycolysis versus gluconeogenesis

PFK1 vs. F 1,6 BPtase1

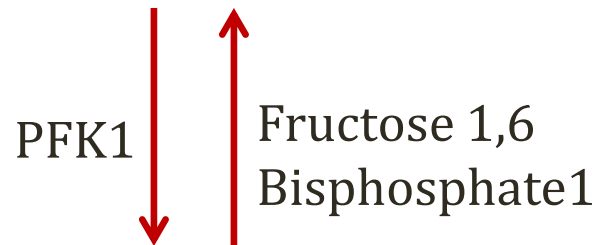
Phosphofructokinase-1
Glycolysis



Fructose 1,6 Bisphosphatase
Gluconeogenesis



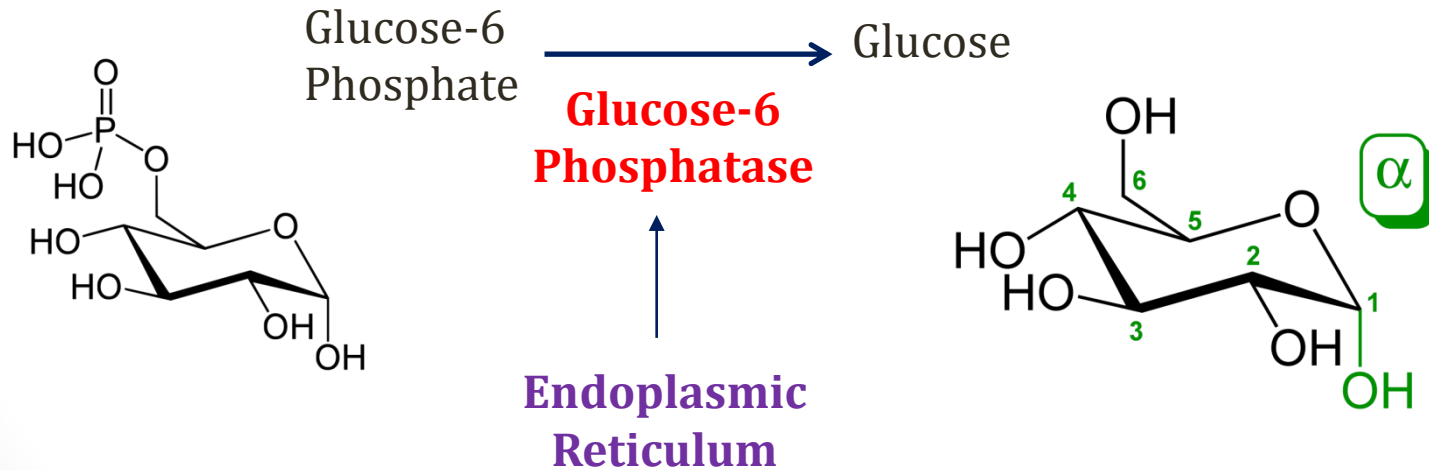
Fructose-6-phosphate



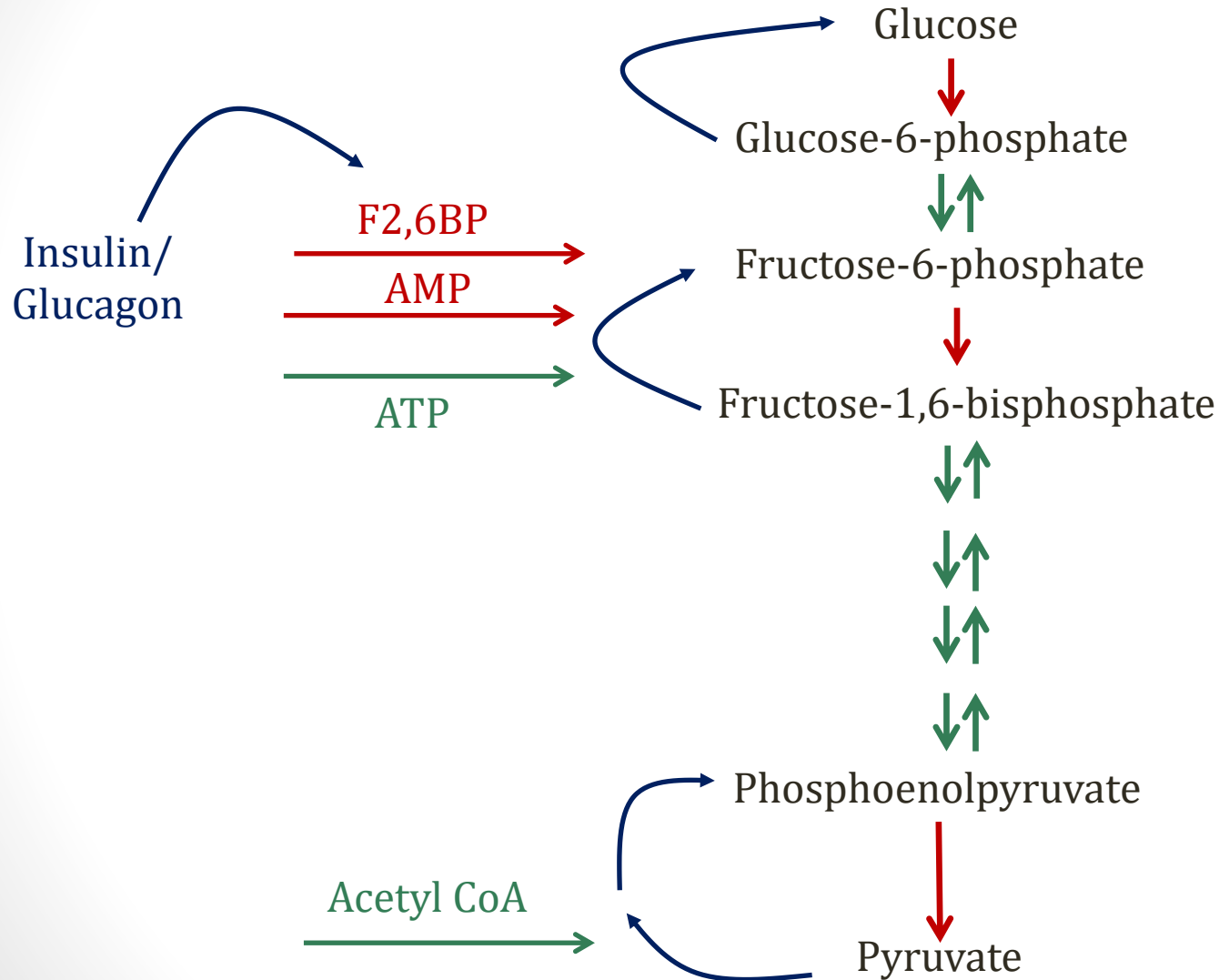
Fructose-1,6-bisphosphate

Gluconeogenesis

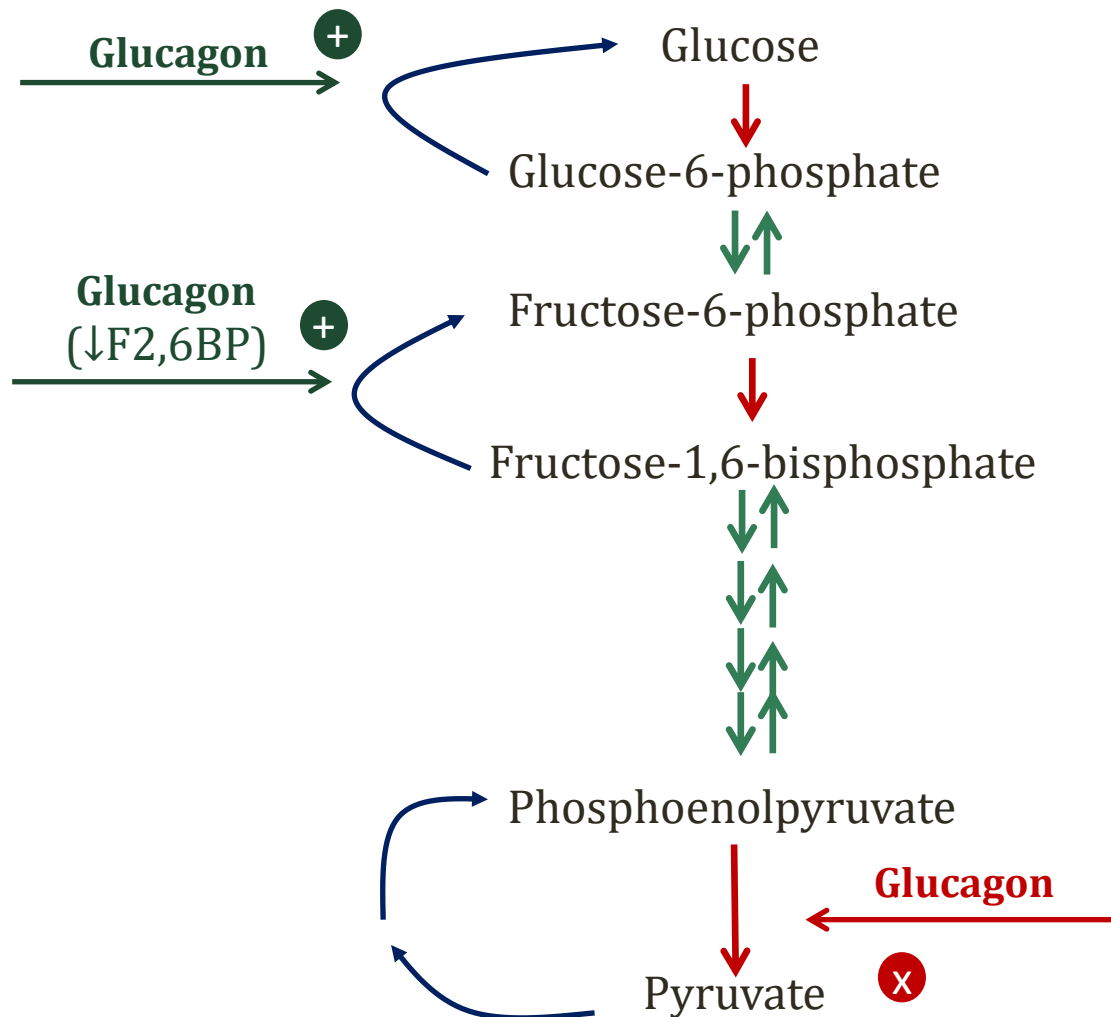
- Step #3: Glucose 6-phosphate → Glucose
- Occurs mainly in liver and kidneys
- Other organs shunt G6P → glycogen



Gluconeogenesis

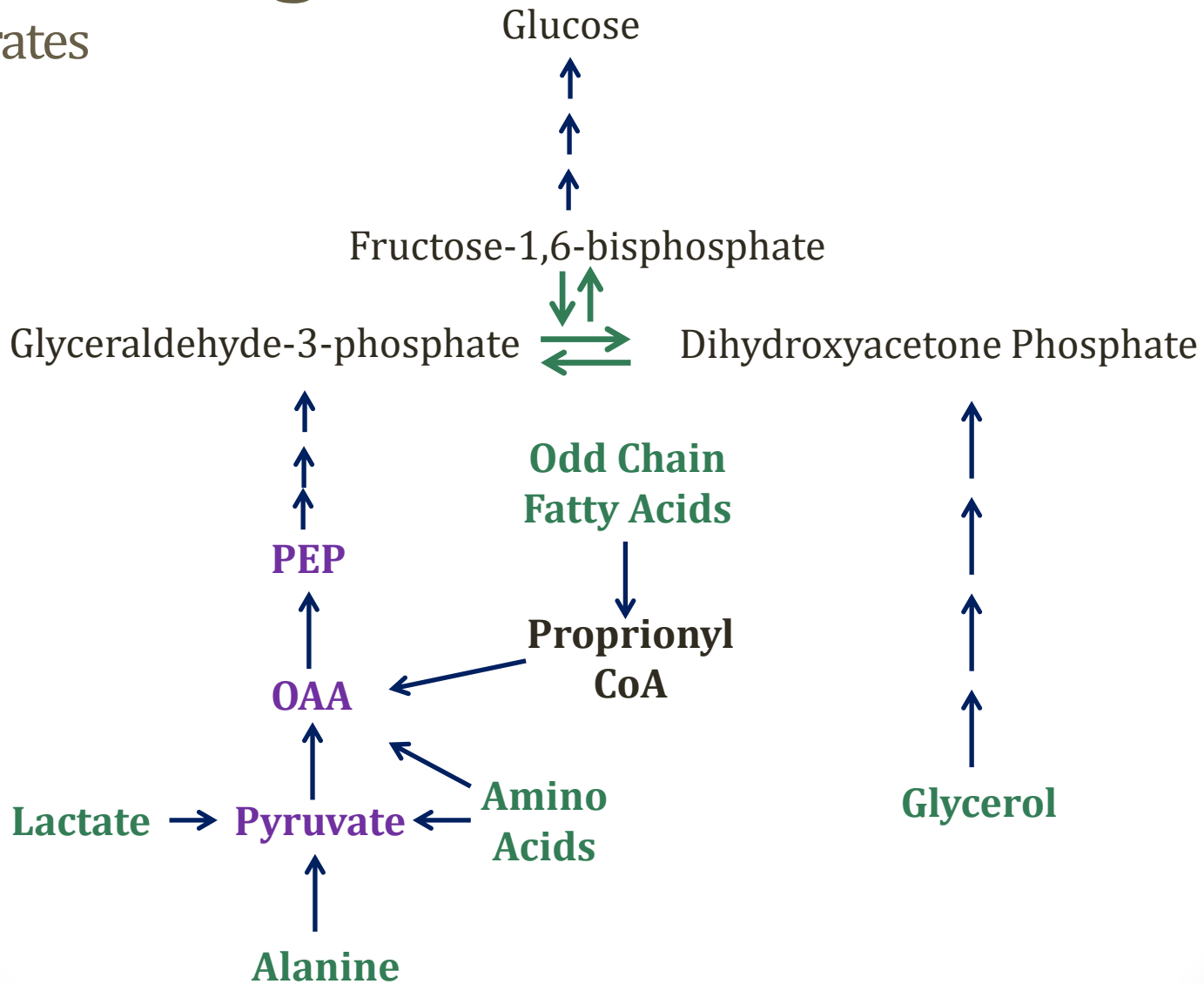


Hormonal Control



Gluconeogenesis

Substrates



Hormones

- Insulin
 - Shuts down gluconeogenesis (favors glycolysis)
 - Action via F 2,6, BP
- Glucagon (opposite of insulin)

Other Hormones

- Epinephrine
 - Raises blood glucose
 - Gluconeogenesis and glycogen breakdown
- Cortisol
 - Increases gluconeogenesis enzymes
 - **Hyperglycemia** common side effect steroid drugs
- Thyroid hormone
 - Increases gluconeogenesis